

Licensed Electrician's Theory (LET) Assessment

Sample Paper Three Marking guide (2020)



AS/NZS 3000 Wiring Rules

Q.1

Protected from mechanical damage likely to occur at the location (1 mark), Clause 5.5.1.2 (c) (1 mark)

Suitably protected against corrosion (1 mark) Clause 5.5.1.2 (d) (1 mark)

Q.2

No (2 marks) Clause 2.3.7.3 (2 marks)

Q.3

230V (Deduct 1 mark for no or incorrect units given) (2 marks)

Clause 4.14.4 (2 marks)

Q.4

230/400 volt. +10% -6% (Deduct 1 mark for no or incorrect units given) (2 marks)

Clause 1.6.2 (c) (a) (2 marks)

AS/NZS 3012 Construction & Demolition Sites.

Q.5

Heavy duty double insulated with a minimum size of minimum 4mm² (Deduct 1 mark for no or incorrect units given) (2 marks)

Clause 2.8.4 (2 marks)

Q.6

(One of the following) To supply power to electrical equipment and lighting immediately adjacent to the structure (2 marks) Clause 2.9 (d) (ii) (2 marks) OR

Only supply power to other transportable structures when the socket outlet is part of an interconnecting system (2 marks), Clause 2.9 (d) (iv) (2 marks)

Electrical Safety (General) Regulations 2019

Q.7

Must not exceed 200mm (2 marks)

Reg 219 (3) (2 marks)



Electrical Shock Survival

Q.8

Any two of the below, or variations of the same. Must include at least one of the 'call for help items (items 1 or 2 below)

1. Dial 000/mobile 112/call emergency services/ambulance (2 marks)
2. If available send for Automatic External Defibrillator (AED) (2 marks)
3. If alone, shout for help (2 marks)
4. Commence first aid (2 marks)

Cable Selection

Q.9

Table 3(3) Item 4 (1 mark)

Table 14 Col 23 (2 marks, one for table and one for column)

Derating factor Table 25(2) Column 2 - 0.81 (1 mark)

Part (i) Answer: 330.48A (2 marks)

Part (ii) Derating factor Table 29 Column 2 – 1.04 (1 mark)

Answer: 343.70A (1 mark)

(Deduct 1 mark for no or incorrect units on final answers. Only deduct one mark regardless of number of missing units)

DC Circuits

Q.10

(i) Current in resistor C = 3.2A (2 marks)

(ii) Voltage across resistor B = 32V (2 marks)

(iii) Power dissipated in resistor D = 432W (2 marks)

(Deduct 1 Mark for no or incorrect units)

Maximum Demand

Q.11

Table C2 Column 3 (1 mark)

Domestic Residence.

Equipment	Load Group	Calculation	Maximum Demand
3 ccts of 10 -10A socket outlets 30 Points total	(b)(i)	$1000W + (29 \times 750W) = 22750W$ $22750 / 230 = 98.91A$	98.91A (2 marks)
1 cct of 2 – 15A socket outlets	(b)(iii)	$15A + (15A \times 0.75) = 26.25A$	26.25A (1 mark)
2 ccts of 12 – incandescent lighting points 24 points	(a)	Full Connected Load $24 \times 60 = 1440W / 230 = 6.26A$	6.26A (2 marks)
Air compressor motor rated at 12.5A	(d)	Full Connected Load for highest 12.5A	12.5A (1 mark)
		Total Maximum Demand	143.92A (1 mark)

(Deduct 1 Mark for no or incorrect units on total, deduct 1 mark for no or incorrect load groups)

Voltage Drop

Q.12

Consumer Mains

Table 43 Column 10 (1 mark)

Vc 1.94 (1 mark)

Vd 15.13V (1 mark)

Sub-mains

Table 45 Column 10 (1 mark)

Vc 2.67 (1 mark)

Vd 4.49V (1 mark)

Final Sub-circuit

Table 45 Column 6 (1 mark)

Vc 4.04 (1 mark)

Vd 3.27V (1 mark)

Total Voltage Drop= $15.13 + 4.49 + 3.27 = 22.89V$ (1 mark)

(Deduct 1 Mark for no or incorrect units on total. Deduct 1 mark for no or incorrect table number/s)

Overload & Short Circuit Calculations

Q.13

Overcurrent divided by MCB current rating = 2 (1 mark)

Minimum Time = Accept 24-30 seconds (1 mark)

Maximum Time = Accept 110-120 seconds (1 mark)

(Deduct 1 mark for no or incorrect time unit)

Q.14

Transformer impedance

230/13,500 (2 marks)

0.01704Ω (1 mark) Answer must be to 5 decimal places.

Main switchboard prospective fault

230/ (0.01704 +0.0047) (2 marks)

10,579.58A (1 mark)

Distribution board prospective fault

230/ (0.01704 + 0.0047 + 0.023) (2 marks)

5140.81A (1 mark)

(Deduct 1 Mark for no or incorrect units in final answer)

Residual Current Devices

Q.15

20A (1 Mark)

AS/NZS 3000 Clause 2.6.2.1 (b) (2 marks)

Motor and Starters

Q.16

D (2 Marks)

AS/NZS 4836:2023

Q.17

Before and after use (2 marks)

Clause number: 8.5.2 (2 marks)

Installation Defects- Non Domestic

Q.18

(2 marks for correct defect one mark for the correct clause)

(Only accept the first 5 defects candidate has listed)

1. Consumers mains not installed in a manner that can maintain supply if exposed to fire: Clause 7.2.2.1
2. Distribution Board Main Switch not labelled with the portion of the installation it controls: Clause 2.3.3.5 (b)
3. DB neutral connections at the Neutral bar are not in the correct sequence with circuit protection: Clause 2.10.5.4
4. Main Switch Fire Pump does not clearly and reliably indicate the on/off position: Clause 2.3.2.2.1 (c)
5. Main Switch Fire Pump is not marked "IN THE EVENT OF FIRE DO NOT SWITCH OFF" : Clause 7.2.4.4 (b)
6. Main Switch DB is not rated for MD current: Clause 2.2.4.3 (a)
7. Location of the earth electrode is not identified on the main switchboard: Clause 5.3.6.4
8. The main earthing conductor is uninsulated: Clause 5.3.2.4
9. DB earthing conductor is undersized for 70mm² active conductors: Clause 5.3.3.1.2 (a)
10. The fire pump MIMS cables are of insufficient current carrying capacity: Clause 3.4.1
11. No verification record on or at the main switchboard: Clause 8.4
- 12.